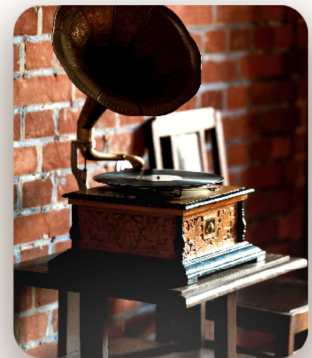


Interactive AR Science Museum

Discover Science with Augmented Reality



Made with Unity

PHOTOGRAPHIC APPARATUS

A TECHNICAL COMPENDIUM OF EARLY PHOTOGRAPHY

Vol. XVII — Institute of Photographic Sciences — Est. 1839

Lens Assembly

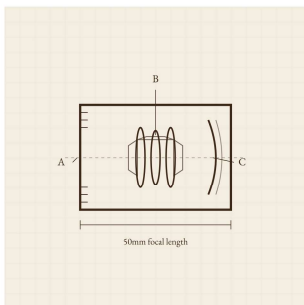
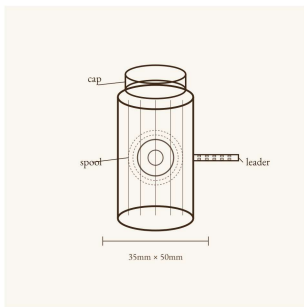


Fig. I — Technical drawing, 1965

APPARATUS PARTS

- A — Lens Element
B — Aperture Iris
C — Focus Ring
D — Mount Thread

35mm Film Canister



Standard 36-exposure roll

Cellulose acetate base with light-sensitive silver halide coating.

Single-Lens Reflex Camera

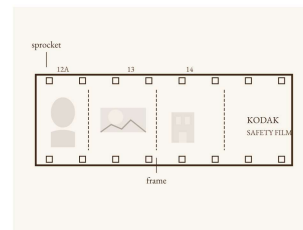
Praktica Series, circa 1960s



§1. This apparatus employs a reflex mirror system whereby light entering through the objective lens is reflected upward to the viewing pentaprism, permitting the operator to compose the image precisely as it shall appear upon the photographic emulsion.

§2. The mechanical shutter mechanism operates at speeds ranging from 1/1000 to multiple seconds, with precision-ground glass elements providing superior optical clarity and minimal chromatic aberration.

Photographic Negative



Developed emulsion

Reversed tonal values for contact printing and enlargement.

Shutter Mechanism

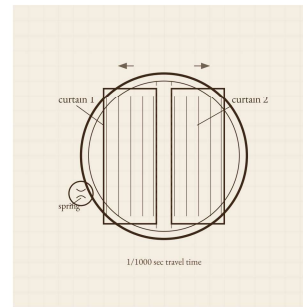


Fig. IV — Technical drawing, 1965

EXPOSURE TABLE

Bright Sun	f/16, 1/125s
Overcast	f/5.6, 1/60s
Interior	f/2.8, 1/30s

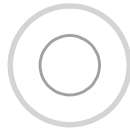
MICROSCOPY

A Study in Optical Science • Est. 1850



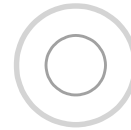
Cell Structure

Detailed examination of plant and animal cellular composition with unprecedented visualization.



Molecular Diagrams

Hand-rendered illustrations documenting molecular arrangements and crystalline formations.



Laboratory Notes

Systematic recordings of experimental observations and scientific hypotheses.



SPECIFICATIONS

Magnification:	40X to 400X
Objective Lens:	Achromatic
Illumination:	Mirror Reflection
Focus System:	Coarse & Fine Adjustment
Stage:	Mechanical, Graduated

OBSERVATIONS

This instrument represents a pivotal advancement in biological and chemical engineering. The brass construction ensures stability and precision. An achromatic lens system provides unparalleled clarity in the observation of minute specimens. Suitable for botanical, zoological, and histological investigations.

EARLY TELECOMMUNICATIONS

Archival Collection • 1876-1920



SPECIMEN NO. TC-1892-07 • CANDLESTICK TELEPHONE

Candlestick Telephone

1890s-1920s

The candlestick telephone represents a pivotal advancement in voice communication technology. Featuring a vertical stand design with separate mouthpiece and earpiece, this device became the standard for early 20th century telephony.

Technical Specifications

Patent Documentation

Carbon microphone transmitter technology enabled clear voice transmission over copper wire networks. The electromagnetic receiver converted electrical signals back into audible speech.

Historical Context

Industrial Revolution Era

By 1892, telephone networks were expanding rapidly across urban centers. This design dominated American and European markets until the introduction of the integrated handset in the 1920s.

Evolution of Voice Communication

1876

First telephone patent granted to Alexander Graham Bell

1892

Candlestick design becomes industry standard

1919

Rotary dial introduced for automatic switching

1927

First transatlantic telephone service established

⚖ EQUILIBRIUM ⚖

A Treatise on Historical Measurement & Balance · Est. MDCCCL



Fig. I — Precision Balance, Brass & Iron Construction, Circa 1850

Technical Specifications

Maximum Capacity: 5 kg

Sensitivity: 0.1 gram

Material: Brass, Iron, Wood

Provenance: European Manufacture

Condition: Excellent

Principles of Equilibrium

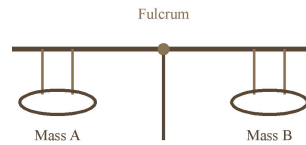


Fig. II — When in equilibrium: $\text{Mass A} \times \text{Distance} = \text{Mass B} \times \text{Distance}$

"The principle of equilibrium remains unchanged since antiquity, wherein equal masses produce equal moments about the fulcrum..."

— Archival Notes, 1852

Museum Archive Collection

CAT-001



Precision Balance

Year: 1850

Origin: France

CAT-002



Apothecary Scale

Year: 1872

Origin: Germany

CAT-003



Merchant's Balance

Year: 1845

Origin: England

THE PHONOGRAPH

A Chronicle of Sound Recording History - Est. 1877



"The phonograph is my baby..." - Edison, 1877

Acoustic Principles

Sound waves are mechanical vibrations captured via diaphragm and stylus, etching grooves into rotating media.

$$\lambda = v / f$$

Recording Mechanics

Groove depth and width correspond to sound amplitude and frequency. Hill-and-dale vs. lateral recording methods.

$$A = A_0 \sin(2\pi ft)$$

Historical Timeline

1857 - Phonautograph
É.-L. Scott de Martinville

1877 - Phonograph
Thomas Edison

1887 - Gramophone
Emile Berliner



Caruso's Arias - 1902



Moonlight Sonata - 1915



Jazz Origins - 1923



Symphony No. 5 - 1910



How to Use

- 1 Download and open the app on your phone
- 2 Press the START button to activate the camera
- 3 Point the camera at any product image in the booklet
- 4 The 3D model will appear directly above the image
- 5 Interact with the model: drag with one finger to rotate it in any direction



Available Products:

Precision Balance Scale

Classic Telephone

Optical Microscope

Photographic Camera

Gramophone



Tips for Best Experience:

Ensure good lighting conditions

Place the booklet on a flat surface

Maintain 20-40 cm distance from camera



Made with Unity